Joseph Wilson

Email: joseph.wilson1@uqconnect.edu.au
Ph: 0400234540,
Website: https://josephwilsonmaths.github.io
Location: Brisbane, Australia

Research Summary

I am a PhD student in Mathematics at The University of Queensland, researching the theoretical foundations of modern AI systems, with a focus on uncertainty quantification and generalization. My work combines mathematical analysis and large-scale empirical evaluation to better understand and improve the reliability of machine learning systems.

Education

July 2023 - Present

PhD, Mathematics - The University of Queensland

- Advisors: Professor Fred Roosta (UQ, CIRES), Dr. Liam Hodgkinson (Uni. Melb), Dr. Chris van der Heide (Uni. Melb).

2019 - 2022

Bachelor of Mathematics (Honours I) - The University of Queensland

- Honours thesis: "A Mathematical Model of Mechanical Cell Competition in Epithelial Tissue". Supervisor: Dr. Zoltan Neufeld.
- GPA 6.53, Dean's Commendation for Academic Excellence, Sem. 1 2022, Sem. 2 2022, Sem. 1, 2023, Sem. 2 2023.

Publications

- *Why Heavy-Tailed Weight Spectra Predict Model Quality*
  Joseph Wilson, Chris van der Heide, Liam Hodgkinson, Zhichao, Wang, Fred Roosta, Michael Mahoney
  Conference on Neural Information Processing Systems (**NeurIPS**) 2026
- *Is the Last Layer Sufficient for Uncertainty Quantification?*
  Joseph Wilson, Chris van der Heide, Liam Hodgkinson, Fred Roosta
  International Conference on Machine Learning (**ICML**), 2026
- *Uncertainty Quantification with the Empirical Neural Tangent Kernel*
  Joseph Wilson, Chris van der Heide, Liam Hodgkinson, Fred Roosta
  Conference on Neural Information Processing Systems (**NeurIPS**) 2025
- *Mechanical Cell Competition in a Model of Epithelial Layer with Size Dependent Proliferation*

Chenhao Zhang, Leo Clarke, **Joseph Wilson**, Faris Saad Alsubaie, Zoltan Neufeld
Mathematical Medicine and Biology: A Journal of the IMA, 42:4, December 2025

Invited Talks and Presentations

- Seminar, QANZIAM 2026, Griffith University, QLD, AU, June 2026
  Is the Last Layer Sufficient for Uncertainty Quantification?
  (awarded best student presentation)
- Seminar, Australian Probability Conference 2026,The University of Queensland, QLD, AU, JUNE 2026
  Is the Last Layer Sufficient for Uncertainty Quantification?
- Seminar, The School of Business Analytics, The University of Sydney, NSW, AU, Feb 2026
  Is the Last Layer Sufficient for Uncertainty Quantification?
- Seminar, The University of California, Berkeley, Berkeley, CA, USA, Dec 2025
  Is the Last Layer Sufficient for Uncertainty Quantification?
- Seminar, Workshop on Optimisation, Metric Bounds, Approximation and Transversality (WOMBAT), The University of Queensland, QLD, AU, Dec 2025
  Uncertainty Quantification with the Empirical Neural Tangent Kernel
- Seminar, 5th Biennial Meeting of the Pacific Northwest Section of SIAM, The University of Washington, Seattle, WA, USA, Oct 2025
  Uncertainty Quantification with the Empirical Neural Tangent Kernel

Employment / Teaching

2022 - Present
Tutor - The University of Queensland

- Courses tutored: MATH1051, MATH1052, STAT2003, STAT2004, STAT2203, STAT3001, STAT3006, MATH3102, MATH3204.

Jan 2023 - July 2023
Control Engineer - Whitelaw Hydraulics

- Software programming, development, and on-site maintenance for control systems for agricultural and industrial applications. Developed AI system for self-navigation, programming languages: CodeSYS, C#.

Service

Jan 2025 - Present
Mathematical Aspects of Machine Learning (MAML) Seminar Series, UQ

Coordinator and co-founder, responsible for finding weekly speakers, notifying attendees of changes, and organizing room bookings.

Professional Development

Jan 2024 - Feb 2024

AMSI Winter School

Completed courses: *Machine Learning/Deep Learning* and *Bayesian Inference and Computations*.